

Mini-Project (ML for Time Series) - MVA 2024/2025

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Introduction

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- ▶ Shapelets are used in time series classification to identify patterns that distinguish different classes.
- ▶ They are **interpretable**, providing meaningful insights as class-discriminative subsequences.
- ▶ This project focuses on implementing a fast version of the brute-force Shapelet algorithm, called **Fast Shapelets**, which leverages SAX representation for efficiency while preserving accuracy.

The Fast Shapelets Algorithm

SAX Representation

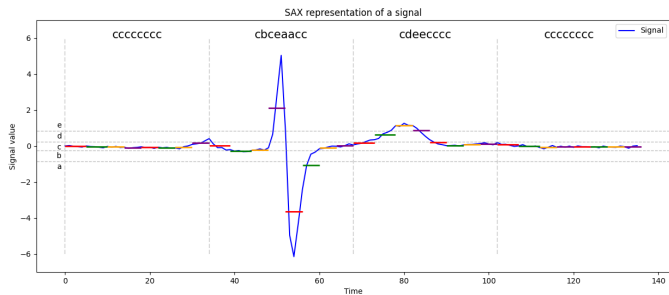


Figure: SAX representation of a signal

Random Projection

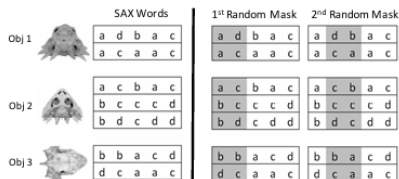


Figure: Random mask (source : ⁰)

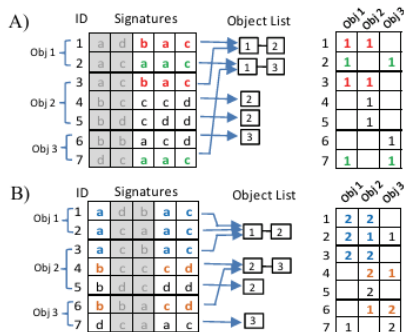


Figure: Collision table (source : ⁰)

Shapelet Selection

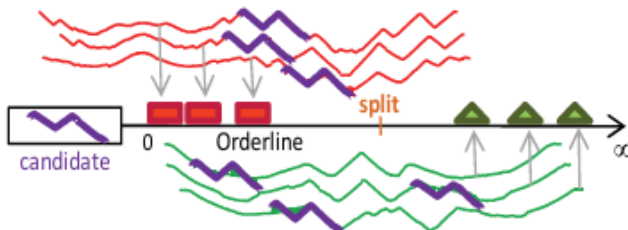


Figure: Shapelet selection (source : ⁰)

⁰T. Rakthanmanon, E. Keogh : *Fast Shapelets: A Scalable Algorithm for Discovering Time Series Shapelets*

Data

Data: ECGFiveDays Dataset

- ▶ The ECGFiveDays dataset contains univariate time series of ECG recordings from a single patient.
- ▶ The dataset has 22 training examples and 861 test examples, leading to an imbalanced setup.
- ▶ Each time series has a fixed length of 136.
- ▶ Classes are globally similar, making them challenging to distinguish.

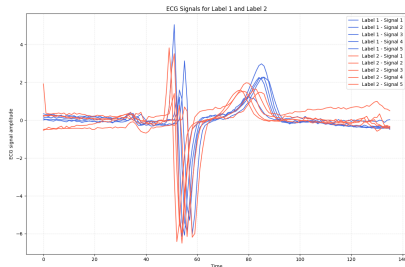


Figure: Example of ECG recordings

Results

Results - Classification

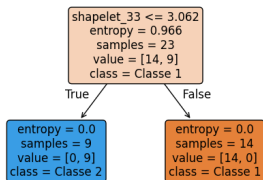


Figure: Classification tree

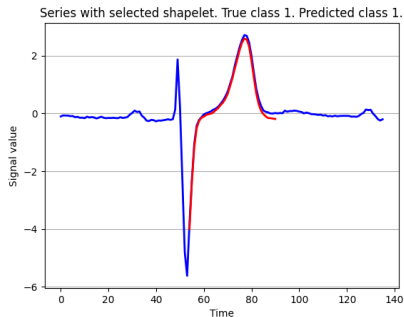


Figure: Signal with selected shapelet

Results - Metrics

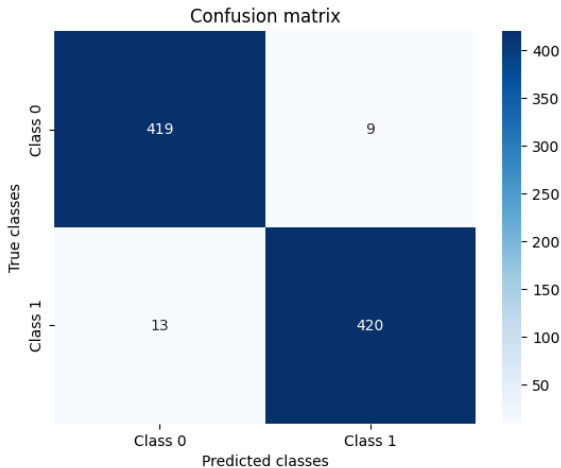


Figure: Confusion matrix

Accuracy: 97.44%

Experiments on Parameters

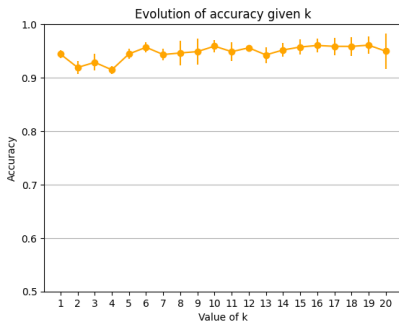


Figure: Accuracy given k

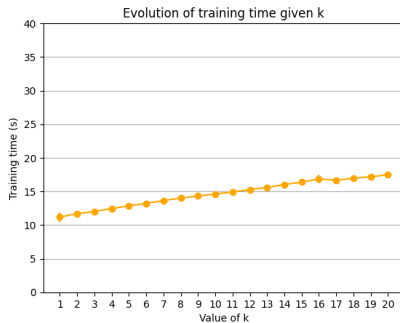


Figure: Training time given k

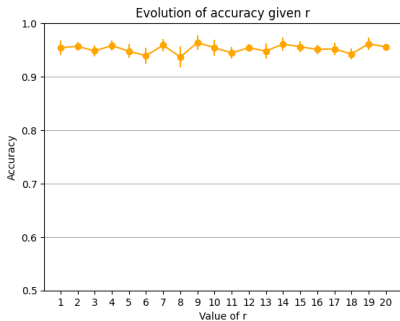


Figure: Accuracy given r

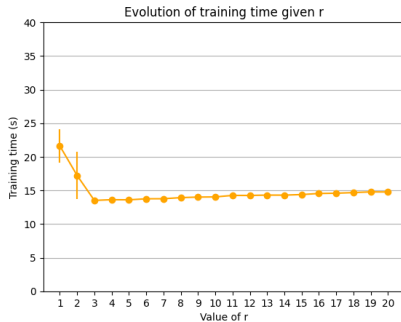


Figure: Training time given r

Dimensionality

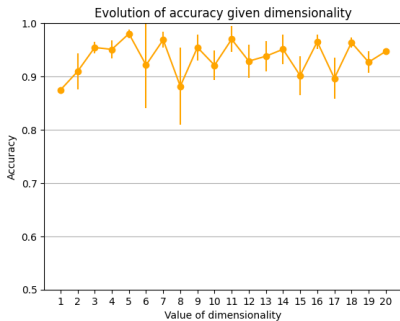


Figure: Accuracy given dimensionality

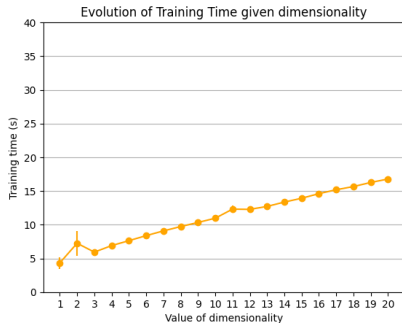


Figure: Training time given dimensionality

Cardinality

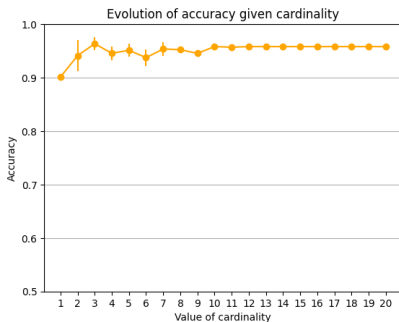


Figure: Accuracy given cardinality

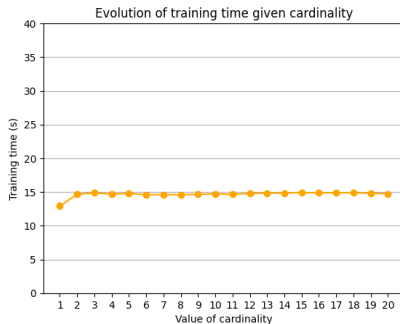


Figure: Training time given cardinality

Mask probability

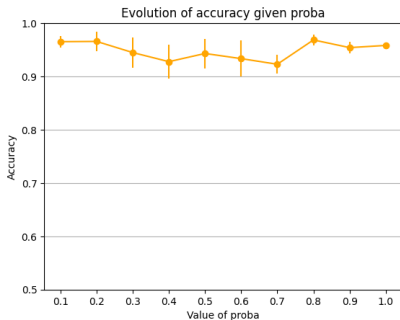


Figure: Accuracy given mask probability

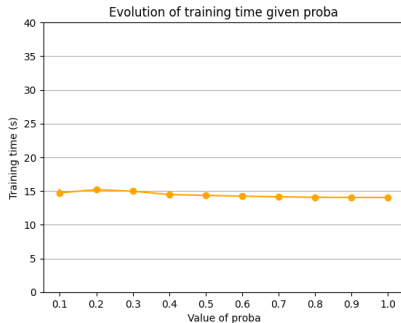


Figure: Training time given mask probability

Limits of Fast Shapelets

Randomness (1/2)

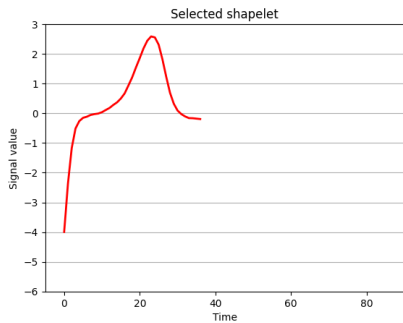


Figure: Shapelet of length 37

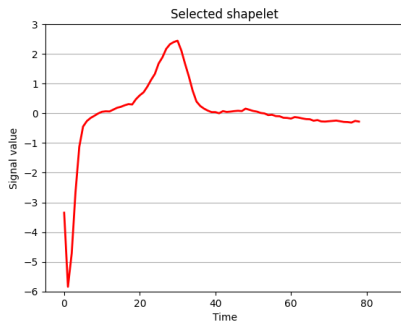


Figure: Shapelet of length 79

Randomness (2/2)

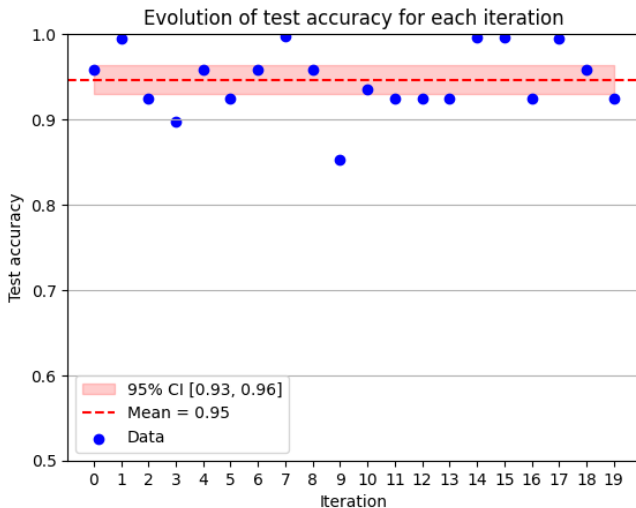


Figure: Accuracy for 20 iterations

Noise in data

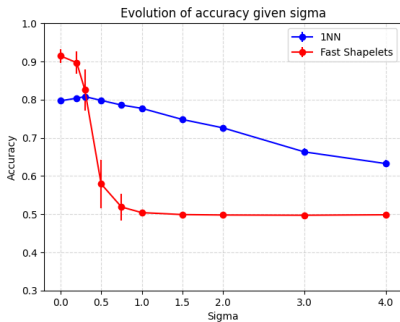


Figure: Accuracy with noise in test data only

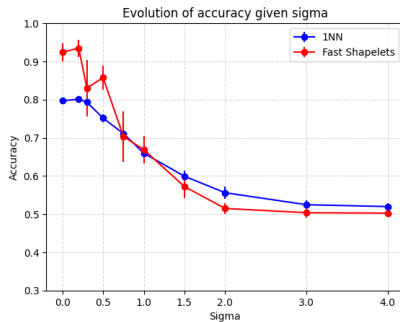


Figure: Accuracy with noise in both train and test data

Conclusion

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- ▶ Fast Shapelets are indeed much faster than the brute force algorithm while maintaining a very high accuracy.
- ▶ Though, in real life scenarios, Fast Shapelets might not be the best choice.
- ▶ Would be great to try those experiments on other datasets, as one dataset is not sufficient to generalize.